Application Programming Interface

Two approaches to APIs

Views provide access

- Views public
- Tables private

Simpler

Functions provide access

- Functions public
- Tables private

More flexible

Either way, table def'ns can change without changing API.

Function has what privileges?

Default: Caller's

Can specify that it uses definer's privileges so that the function provides “public” access to restricted implementation.

```
CREATE FUNCTION InsertEmployee (
    emp_name TEXT,
    emp_salary INT,
    emp_hire_date DATETIME)
RETURNS VOID
AS $$
    DECLARE
        matches INT;
    BEGIN
        SELECT COUNT(*) INTO matches
        FROM Employee
        WHERE name = emp_name);
        IF matches = 0
            INSERT INTO Employee VALUES
             (emp_name, emp_salary, emp_hire_date);
        END IF;
    END $$
LANGUAGE plpgsql
SECURITY DEFINER;
```

View has what privileges?

```
CREATE VIEW patient_restricted(id, name, address)
WITH (security_invoker=TRUE)
AS SELECT id, name, address
   FROM Patient
```

Default: View owner

Can specify that it uses caller's privileges so that the function **does not** provide "public" access to restricted implementation.